Propeller Indicating – General Data

1. General

The propeller control systems, incorporate and provide the following indications:

- Propeller speed in percent rpm
- Propeller reset caution
- Beta range indicting
- Propeller autofeather select and arm

A. Propeller Tachometer Indication

A tachometer mounted on the reduction gearbox provides a signal that is translated by the DAU into a % RPM indication (N_P) which is displayed as part of the engine pane on each PFD.

B. Propeller Reset Caution Light

An amber CAS message with the text Reset Props will appear if the power levers are retarded below the 75% N_G position with the lever set at 91% N_P or less. This caution message indicates the need to move the propellers forward to full INCREASE in order to put the propeller governor into an underspeed condition and provide the pilot with direct control of propeller blade angle via the power levers. The Reset Props message electrical circuit is energized through two microswitches connected in series, one operated by the left propeller lever and the other by the right power lever. The circuit is powered from the right DC bus and is protected by the PROPELLER OVERSPEED GOVERNOR circuit breaker on the main circuit breaker panel.

C. Beta Range Indicating Lights

The beta backup system has been deleted from the Series 400 Twin Otter. There is Indication that a propeller is in ground fine range (blade angles of +9° or less) during normal operations (power lever grips twisted). If a propeller enters ground fine range and the power lever grip switch has not been twisted, a warning level message Prop Low Pitch will be displayed, and an aural alert PROPELLER LOW PITCH will be voiced, in addition to the indications shown above.

Correct operation of the propeller position sensing microswitches can be confirmed by observing that the cyan BETA annunciation appears when the power lever grips are twisted and moved aft. Correct functioning of the propeller lever grip switch can be confirmed by observing the appearance of the status CAS message Interlock Open whenever the power lever grips are twisted, and confirming that the Interlock Open message extinguishes when the power lever grips are no longer twisted.

D. Propeller Autofeather Indicator Lights

Refer to 61-22-02, Propeller Autofeather System (Mod 6/2022) – General Data.

This Page Intentionally Left Blank